

SAT Hard Question Set 14

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Phase 2 · Hard Question Practice

Overview

1 Hard Question Set 14

Question 1

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$ respectively. What is the value of x ?

Answer 1

Final Answer: 5

Question 2

If a and b are numbers greater than 1 and $\sqrt[7]{a^6}$ is equivalent to $\sqrt{b^3}$, for what value of n is $a^{2n+1} = b$?

Answer 2

Final Answer: $\boxed{-\frac{3}{14}}$

Question 3

The function c gives the partial length, in centimeters (cm), of the circumference of a circle with radius r cm:

$$c(r) = \frac{2}{5}(2\pi r)$$

For each increase of 18 cm in r , this partial length increases by $k\pi$ cm. What is the value of k ?

Answer 3

Final Answer: $\boxed{\frac{72}{5}}$

Question 4

For each real number r , which of the following points lies on the graph of both equations in the xy -plane for the given system?

$$15x + 45y = 33$$

$$5x + 15y = 11$$

- A. $(\frac{11r}{5} - 3, r)$
- B. $(-3r + \frac{11}{5}, r)$
- C. $(r, -\frac{r}{3} - \frac{11}{15})$
- D. $(r, \frac{r}{3} + \frac{11}{15})$

Answer 4

Correct Answer: B

Question 5

If $\frac{a}{b} = \frac{4}{5}$ and $\frac{5b}{ta} = 35$, what is the value of t ?

Answer 5

Final Answer: $\boxed{\frac{5}{28}}$

Question 6

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c ?

$$\frac{1}{2cx} = \frac{x}{16} + \frac{1}{c}$$

Answer 6

Final Answer: -8

Question 7

The given equation is one equation in a system of two linear equations. If the system has at least one solution, which of the following equations could be the other equation in the system?

$$7x + 5y = 2$$

I $10.5x + 7.5y = 3$

II $10.5x - 7.5y = 3$

A. I only

B. II only

C. I and II

D. Neither I nor II

Answer 7

Correct Answer: C

Question 8

The function f is defined by $f(x) = a^x - b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ passes through the points $(c, 7)$ and $(2c, 117)$, where c is a constant. Which of the following could be the value of b ?

Answer 8

Final Answer: 4

Question 9

A right rectangular prism has a base area of $16t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the prism is 9 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

Answer 9

Final Answer: $140t + 48$

Question 10

The mass of object A is 498% of the mass of object B, and the mass of object A is 0.830% of the mass of object C. If the mass of object C is $p\%$ of the mass of object B, what is the value of p ?

Answer 10

Final Answer:

Question 11

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

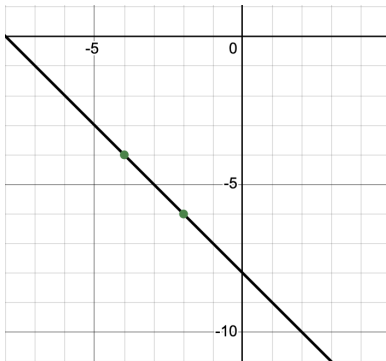
- A. $3x^2 + 7x + 14b$
- B. $3x^2 + 16x + 14b$
- C. $3x^2 + 18x + 14b$
- D. $3x^2 + 25x + 14b$

Answer 11

Correct Answer: D

Question 12

The graph of the linear function $y = f(x) - 19$ is shown. If c and d are positive constants, which equation could define f ?



- A. $f(x) = d - cx$
- B. $f(x) = -d + cx$
- C. $f(x) = d + cx$

Answer 12

Correct Answer: A

Question 13

The expression kx represents the result of decreasing a positive force, x , acting on an object by 0.34%. What is the value of k ?

- A. 0.34%
- B. 1.34%
- C. 99.66%
- D. 100.34%

Answer 13

Correct Answer: C

Question 14

A regular polygon has exactly 87 sides. If the measure of each of the 87 interior angles of this polygon is $(180p)^\circ$, what is the value of p ?

- A. $\frac{85}{87}$
- B. $\frac{37}{87}$
- C. $\frac{87}{90}$
- D. $\frac{87}{180}$

Answer 14

Correct Answer: A

Question 15

The function $f(n) = 7(20.44)^{\frac{n}{4}}$ gives each term of a sequence as a function of the term's position n in the sequence, where n is a whole number. The value of the term in position 14 is $p\%$ more than the value of the term in position 10. What is the value of p ?

Answer 15

Final Answer: 1944

Question 16

A right square pyramid has a total surface area of 28,160 square inches, and the combined surface area of the four lateral faces of this pyramid is 16,060 square inches. What is the height, in inches, of this pyramid?

- A. 48
- B. 55
- C. 73
- D. 110

Answer 16

Correct Answer: A

Question 17

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

$$\sqrt{k - x} = 32 - x$$

Answer 17

Final Answer: 127

Thank You!

We appreciate your time and focus.

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